1 **(a)** $\tan 60° = \dfrac{h}{x}$ giving $x = \dfrac{h}{\sqrt{3}}$ **[M1]**

$\tan 30° = \dfrac{h}{x+12}$ giving $\dfrac{1}{\sqrt{3}} = \dfrac{h}{\frac{h}{\sqrt{3}}+12}$ **[M1]**

$\dfrac{h}{\sqrt{3}} + 12 = h\sqrt{3}$ multiplying by $\sqrt{3}$ gives $h + 12\sqrt{3} = 3h$ **[M1]**

$2h = 12\sqrt{3}$ giving $h = 6\sqrt{3}$ m **[A1]**

(b) $x = 6$ m and $PR = \dfrac{6\sqrt{3}}{\sin 60°} = 12$ m **[B1]**

$PQ = \dfrac{18}{\cos 30°} = \dfrac{36}{\sqrt{3}} = 12\sqrt{3}$ m **[M1]**

Perimeter $= 12 + 12 + 12\sqrt{3} = 24 + 12\sqrt{3}$ **[A1]**

(c) $\dfrac{1}{2} \times PR \times d = \dfrac{1}{2} \times RT \times PT$ (or $d = 6\sin 60°$) **[M1]**

$d = \dfrac{6 \times 6\sqrt{3}}{12}$ **[M1]**

$3\sqrt{3}$ **[A1]**

[Total Marks: 10]

2 Identifies cosine ratio: $\cos(\theta) = \dfrac{2.1}{6.5}$ **[M1]**

$\theta = \cos^{-1}\left(\dfrac{2.1}{6.5}\right)$ **[M1]**

71.2 **[A1]**

[Total Marks: 3]

3 $\cos(58°) = \dfrac{9.4}{XZ}$ **[M1]**

$XZ = \dfrac{9.4}{\cos(58°)}$ **[M1]**

17.7 or 17.73... **[A1]**

[Total Marks: 3]

4 **(a)** $BC^2 + 8.5^2 = 14.2^2$ **[M1]**

$BC = \sqrt{14.2^2 - 8.5^2}$ OR $BC = \sqrt{129.39}$ **[M1]**

11.4 (accept 11.37...) **[A1]**

(b) $\cos(\angle BAC) = \dfrac{8.5}{14.2}$ **[M1]**

53.2 (accept 53.23...) **[A1]**

[Total Marks: 5]

5 **(a)** Uses tangent ratio: $\tan(34°) = \dfrac{AB}{14}$ or $AB = 14\tan(34°)$ **[M1]**

9.44 (accept 9.443) **[A1]**

(b) Recognises shortest distance is perpendicular to AC, so $\angle BDC = 90°$ **[B1]**

Uses sine ratio in right-angled triangle BDC: $\sin(34°) = \dfrac{BD}{14}$ **[M1]**

7.83 (accept 7.828 to 7.83) **[A1]**

[Total Marks: 5]

6 **(a)** $\tan(38°) = \dfrac{TB}{45}$ **[M1]**

35.2 (or 35.15 to 35.16) **[A1]**

(b) $\cos(38°) = \dfrac{45}{PT}$ or $PT^2 = 45^2 + 35.15^2$ **[M1]**

57.1 (or 57.10 to 57.11) **[A1]**

(c) $\tan(XPB) = \dfrac{20}{45}$ **[M1]**

Angle $= \tan^{-1}\left(\dfrac{20}{45}\right)$ **[M1]**

24.0 (or $23.96\ldots$) **[A1]**

[Total Marks: 7]

7 **(a)** $\tan 54° = \dfrac{AB}{6.5}$ giving $AB = 6.5\tan 54°$ $(= 8.946\ldots)$ **[M1]**

8.95 **[A1]**

(b) $BD = 6.5 + 8.2 = 14.7$ m **[B1]**

$\tan\angle ADB = \dfrac{8.946\ldots}{14.7}$ $(= 0.6086\ldots)$ **[M1]**

31.3 **[A1]**

(c) Shortest distance $d = 14.7\sin 31.32\ldots°$ (or $AD = \sqrt{8.946\ldots^2 + 14.7^2} = 17.208\ldots$ m) **[M1]**

$d = 7.642\ldots$ (or $\dfrac{8.946\cdots \times 14.7}{17.208\ldots}$) **[M1]**

7.64 **[A1]**

[Total Marks: 8]

8 **(a)** Uses $\sin(52°) = \dfrac{\text{distance}}{18}$ or $18 \times \sin(52°)$ **[M1]**

14.18... or 14.2 **[A1]**

(b) Distance to closest point $= 18 \times \cos(52°) = 11.082...$ km **[M1]**

Time $= \dfrac{11.082...}{14}$ $(= 0.7915...$ hours) **[M1]**

$0.7915... \times 60 = 47.49... = 47$ **[A1]**

[Total Marks: 5]

9 (a) $\tan(38°) = \dfrac{h}{50}$ **[M1]**

$h = 50\tan(38°)$ OR $39.06\ldots$ **[A1]**

Total height $= 39.06\cdots + 1.6 = 40.7$ (accept $40.66\ldots$) **[A1]**

(b) Identifies the angle of elevation of T from X as $25°$ OR $\angle BTX = 65°$ **[B1]**

$\tan(25°) = \dfrac{40.66}{BX}$ OR $\tan(65°) = \dfrac{BX}{40.66}$ **[M1]**

$BX = 87.2$ (accept $87.19\ldots$) **[A1]**

[Total Marks: 6]

10 (a) Uses $\angle TPB = 42°$ (or $\angle BTP = 48°$): $\tan(42°) = \dfrac{65}{BP}$ or $BP = 65\tan(48°)$ **[M1]**

72.2 (accept 72.18 to 72.2) **[A1]**

(b) Uses $\angle TQB = 27°$ (or $\angle BTQ = 63°$): $\tan(27°) = \dfrac{65}{BQ}$ **[M1]**

Calculates $BQ = \dfrac{65}{\tan(27°)} = 127.57...$ **[A1]**

Subtracts distances: $PQ = 127.57... - 72.189...$ **[M1]**

55.4 (accept 55.38 to 55.4) **[A1]**

[Total Marks: 6]

11 (a) Calculates $\angle PAB = 360° - (180° - 40°) - 130° = 90°$ (or similar valid reasoning to show right angle) **[M1]**

Uses Pythagoras' theorem: $PB^2 = 80^2 + 60^2$ **[M1]**

100 **[A1]**

(b) Uses trigonometry to find $\angle APB$: $\tan(APB) = \dfrac{60}{80}$ **[M1]**

$\angle APB = \tan^{-1}(0.75) = 36.869...°$ **[A1]**

Bearing $= 040° + 36.869...° = 076.9$ (accept 077) **[A1]**

(c) Recognises the shortest distance d is the perpendicular height of $\triangle PAB$ from A to PB. **[M1]**

Uses area equation: $\dfrac{1}{2} \times 100 \times d = \dfrac{1}{2} \times 80 \times 60$ OR uses trigonometry $80\sin(53.13°)$ or $60\sin(36.87°)$ **[M1]**

48 **[A1]**

[Total Marks: 9]

12 **(a)** Interior angle between parallel North lines $= 180° - 52° = 128°$ **[M1]**

Angles around point $B = 360° - 128° - 142° = 90°$ with no errors seen **[A1]**

(b) $AC = \sqrt{15^2 + 20^2}$ **[M1]**

$AC = \sqrt{625}$ **[M1]**

25 **[A1]**

(c) $\tan(BAC) = \dfrac{20}{15}$ **[M1]**

Angle $BAC = 53.13°$ **[A1]**

Bearing $= 052° + 53.13°$ **[M1]**

105.1 (or $105.13\ldots$) **[A1]**

[Total Marks: 9]

13 **(a)** Bearing of P from $L = 058° + 180° = 238°$ **[B1]**

Angle $PLQ = 238° - 148° = 90°$ **[A1]**

(b) **(i)** $PQ^2 = 24^2 + 32^2 = 576 + 1024 = 1600$ **[M1]**

40 **[A1]**

(ii) $\tan \angle LPQ = \dfrac{32}{24}$ giving angle $LPQ = 53.130\ldots°$ **[M1]**

Bearing $= 058° + 53.130\ldots°$ **[M1]**

111.1 **[A1]**

(c) $\dfrac{1}{2} \times 40 \times d = \dfrac{1}{2} \times 24 \times 32$ (or $d = 24 \sin 53.130\ldots°$) **[M1]**

19.2 **[A1]**

[Total Marks: 9]

14 **(a)** $\angle APL = 90° - 65° = 25°$ **[B1]**

$\cos(25°) = \dfrac{PL}{120}$ OR $\sin(65°) = \dfrac{PL}{120}$ **[M1]**

109 or 108.8 (accept $108.75\ldots$) **[A1]**

(b) Identifies the shortest distance as the perpendicular distance from L to PA, creating a new right-angled triangle with hypotenuse PL **[B1]**

$\sin(25°) = \dfrac{d}{108.76}$ OR Area method: $0.5 \times 120 \times d = 0.5 \times 108.76 \times (120 \sin 25°)$ **[M1]**

46.0 (accept $45.96\ldots$) **[A1]**

[Total Marks: 6]

15 **(a)** $PR = \sqrt{8.5^2 + 11.2^2}$ or $\sqrt{197.69}$ **[M1]**

14.06... or 14.1 **[A1]**

(b) $\sin(41°) = \dfrac{14.06...}{PS}$ or $PS = \dfrac{14.06...}{\sin(41°)}$ **[M1]**

21.43... or 21.4 **[A1]**

(c) Angle $RPS = 180° - 90° - 41° = 49°$ AND distance $= 14.06... \times \sin(49°)$
OR area method: $0.5 \times 21.43... \times h = 0.5 \times 14.06... \times (14.06.../\tan(41°))$ **[M1]**

10.60... or 10.6 **[A1]**

[Total Marks: 6]

16 (a) Sets up equation for $\triangle BCD$: $\tan(42°) = \dfrac{CD}{BC}$

BC $= \dfrac{CD}{\tan(42°)}$ [M1]

Sets up equation for $\triangle ACD$: $\tan(25°) = \dfrac{CD}{BC + 50}$

BC + 50 $= \dfrac{CD}{\tan(25°)}$ [M1]

Equates or substitutes to eliminate BC: $50 = \dfrac{CD}{\tan(25°)} - \dfrac{CD}{\tan(42°)}$ **[M1]**

Factorises to solve for CD: $CD\left(\dfrac{1}{\tan(25°)} - \dfrac{1}{\tan(42°)}\right) = 50$ **[M1]**

48.4 (accept 48.36...) **[A1]**

(b) Recognises the shortest distance d is the perpendicular from B to AD, forming a right-angled triangle with hypotenuse $AB = 50$. **[M1]**

Sets up trigonometric ratio: $\sin(25°) = \dfrac{d}{50}$ **[M1]**

21.1 (accept 21.13...) **[A1]**

[Total Marks: 8]

17 (a) $\angle CAB = 90° - 58° = 32°$ **[B1]**

$\angle CBA = 315° - 270° = 45°$ **[B1]**

(b) Expresses AD and BD in terms of CD: $AD = \dfrac{CD}{\tan(32°)}$ and $BD = \dfrac{CD}{\tan(45°)} = CD$ **[M1]**

Sets up sum equation: $\dfrac{CD}{\tan(32°)} + CD = 48$ **[M1]**

Factors out CD: $CD\left(\dfrac{1}{\tan(32°)} + 1\right) = 48$ giving $CD(2.6003...) = 48$ **[M1]**

18.5 (accept 18.45 to 18.46) **[A1]**

(c) Uses sine ratio in triangle ACD: $\sin(32°) = \dfrac{18.459...}{AC}$ **[M1]**

34.8 (accept 34.83 to 34.9) **[A1]**

[Total Marks: 8]

18 **(a)** $AC = \sqrt{8^2 + 15^2}$ **[M1]**

17 **[A1]**

(b) $AD^2 + 12^2 = 17^2$ (or their AC^2) **[M1]**

$AD = \sqrt{289 - 144} = \sqrt{145}$ **[M1]**

12.0 (or 12.04...) **[A1]**

(c) Area of $\triangle ADC = \frac{1}{2} \times 12 \times \sqrt{145} = 72.25\ldots$ **[M1]**

$\frac{1}{2} \times 17 \times h = 72.25\ldots$ **[M1]**

8.50 (or 8.499...) **[A1]**

(d) $\tan(BAC) = \frac{15}{8} \implies$ Angle $BAC = 61.93°$ **[M1]**

$\tan(CAD) = \frac{12}{\sqrt{145}} \implies$ Angle $CAD = 44.90°$ **[M1]**

Angle $DAB = 61.93° + 44.90°$ **[M1]**

106.8 (or 106.82 to 106.83) **[A1]**

[Total Marks: 12]